Fairness or Bias of Large Language Models (LLMs) in High-Stake Domains such as Hiring and Recidivism

Sudha Sree
MSCS at UIC
Chicago, USA
syerr7@uic.edu

Prathik Pugazhenti
MSCS at UIC
Chicago, USA
ppuga@uic.edu

Meetkumar Raychura
MSCS at UIC
Chicago, USA
mrayc@uic.edu

Abstract

In this paper, we review and compare ten peer-reviewed research papers that investigate fairness and bias in LLMs, with an emphasis on their effect in sensitive high-stake domains such as hiring and recidivism. We have analyzed the papers in terms of four core aspects, which are the methodologies used to test for bias in the papers, the bias detection techniques, bias mitigation strategies evaluation metrics used to check whether the bias has mitigated after applying the bias mitigation techniques. The analysis shows wide differences in how fairness is defined across the papers. Some papers focus on demographic parity, others rely on counterfactual fairness, and many papers focus on other kinds of biases beyond the typical gender, race, and socioeconomic status biases that are talked about. We grouped the papers based on the types of bias they detect. The survey then examines the different evaluation approaches and the bias mitigation methods proposed in these works. Overall, the survey identifies the large number of bias detection techniques that exist and points out the insufficiency of standardized evaluation and transparent documentation. It is imperative that there is such a standardized method of evaluation for LLM applications, especially since they can influence major societal outcomes of the persons being evaluated by them.

CCS Concepts

• **Computing Methodologies** → **Large Language Models in Machine Learning**; • **Machine Learning** → *Fairness and Ethics of Machine Learning*; • **Human-centered computing** → Fairness, accountability, and transparency.

Keywords

Large Language Models, fairness, bias, hiring, hiring tools, recidivism, discrimination, ChatGPT, Gemini, Llama

ACM Reference Format:
Sudha Sree, Prathik Pugazhenti, and Meetkumar Raychura. 2018. Fairness or Bias of Large Language Models (LLMs) in High-Stake Domains such as Hiring and Recidivism. In *Woodstock '18: ACM Symposium on Neural Gaze Detection, June 03–05, 2018, Woodstock, NY*. ACM, New York, NY, USA, 11 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Declaration of AI Usage

We acknowledge that we did not use any AI tools in the submission of this assignment beyond grammar checks and fluency.

2 Introduction

Large Language Models (LLMs) such as ChatGPT and BERT are becoming more common in critical decision making domains like hiring and recidivism due to the benefit they offer of being able to efficiently handle large amounts of data in a short amount of time. Since these LLMs train on data pulled from the internet, the biases in the source are often reflected in the dataset. In hiring decisions, biased model outputs can skew opportunities against particular groups, and potentially contribute to widening the gap between the rich and the poor/unemployed. In recidivism prediction, unfair scoring can reinforce long-standing systemic inequities against certain subgroups of people. As LLMs are becoming increasingly commonly used in high-stakes areas, researchers have started examining different definitions of fairness and develop ways to address these. In this paper, we have summarized the findings from ten peer-reviewed research papers and have grouped them based on the bias they detect. We then assess the practicality and effectiveness of the evaluation and mitigation approaches that are used in these papers in real world scenarios across high-impact domains.

3 Motivation

As LLMs are being used more often in important decision processes, they give considerable power to systems that might not be transparent, accountable or fair enough. When a model's decision affects whether someone gets a job interview or judged as "high risk" in criminal justice, the stakes become deeply personal, with consequences that can be irreversible. High-stakes situations like these demand a stricter oversight well beyond everyday LLM uses. However, the adoption of LLMs has grown so quickly that fairness guidelines haven't kept up. This imbalance between rapidly advancing capabilities and limited governance is what motivates our study.

This problem is particularly striking because there is no consistent or universal understanding of what "fairness" should be defined as across various studies, domains, and assessment methods that could help everyone. Because researchers use different fairness definitions, detection/testing approaches and mitigation techniques, it becomes difficult to interpret any meaningful comparisons or to determine if the model is safe enough for real-world applications. The absence of standardized evaluation framework has resulted in scattered progress, leaving practitioners with little information on how to responsibly apply LLMs to high-stakes situation. We seek to contribute by compiling insights from ten peer-reviewed papers and mapping them into a clear framework that highlights how bias is detected, how fairness is evaluated, and mitigation in real use cases to point out the absence of a unified representation or marking of what fairness is. By combining the findings from these papers, our survey shows what areas in the field still remains fragmented and also stresses how important it is to standardize fairness benchmarks and transparent documentation for high-stakes LLM applications.

4 Technical

4.1 Technical Challenges We Faced

One of the major challenges that we encountered was the lack of clear and transparent dataset descriptions in many of our reviewed papers, which made it difficult for us to evaluate their methodologies. Important information about the datasets from the creation and pre-processing flows to demographic distribution, annotation procedures and presence of societal biases, was often incomplete. Some papers lacked even the most fundamental details, like sampling methods or label creation. Without adequate information, it was difficult for us understand context of dataset, as a result, difficult to determine how much of the observed bias were rooted in the data itself. We also faced problems due to inconsistent reporting of bias metrics and evaluation procedures in the papers. Some papers created their own fairness metrics that did not align with the standard definition while others used domain-specific datasets, which made it difficult for us to comparison their results. To overcome this challenge and to make sense of these terms, we created our own criteria for grouping and interpreting these metrics. Lastly, combining the results from these studies also implied dealing with lot of uncertainty about how each study understood and used the idea of causality. Many papers offered possible reasons of bias, ranging from imbalanced datasets to incorrect correlation patterns and misalignment of training objective. However, only few offered concrete explanations behind these biases. Across several papers, we couldn't clearly tell whether the biases were driven by model features, imbalances in data distribution, filters in pre-processing pipeline or underlying societal pattern reflected in the dataset. Our survey aimed to analyze bias across multiple domains like classification, ranking and LLM outputs. Using what we learned in class about how causal thinking can expose biases not evident at the surface level. The absence of causal clarity across papers made it difficult for us to generalize the findings or draw domain-independent conclusions.

4.2 Vision of Technical Contribution

We started the project with a structured examination of each paper and identifying the methods each paper has used to detect or analyze bias in LLMs. in order to be able to address our Research Question 1 (RQ1), centered on identifying how LLMs demonstrate bias when deployed in hiring and recidivism domains. For each paper, we examined the approaches each study used, ranging from demographic analysis to perturbation-based techniques, ranking-fairness metrics and custom evaluation setup. Next, we focused on the different kinds of bias each research paper focused on and then categorized the findings accordingly. Some of the papers worked on examining representational bias such as stereotyping or sentiment skew. While others addressed allocative bias in high-stakes domains like hiring, admissions and recidivism assessment, which involved critical decision-making tasks. By grouping the papers by their primary bias focus helped us see patterns across different domains and offered key insights towards answering Research Question 2 (RQ2), which focuses on existing bias detection and mitigation techniques.

The next step included analyzing mitigation strategies such as dropout-based fairness repair, dataset rebalancing, adversarial training, fairness-constrained optimization, and post-processing adjustments to outputs or rankings. It helped to address Research Question 3 (RQ3: How can we measure how fair LLM is in high-stakes situations?). We examined how each paper approached fairness evaluation in high-stakes settings and extracted any metrics, frameworks, or procedures that could be generalized. Here, we noted that studies varied drastically in the metrics they prioritized; some used error-rate parity or equalized odds, others used exposure fairness, and many relied on custom formulations that lacked broader applicability. By compiling these findings, our survey highlights not only the methodological diversity but also the conceptual disagreements concerning what "fairness" should mean in high impact contexts. Overall, our technical contribution lies in synthesizing these fragmented approaches and revealing a larger structural issue: the field lacks a standardized, universal evaluatory framework for assessing fairness in LLMs that anyone, even a layman could understand transparently and use. Such a framework would not only support reproducibility and transparency but also empower broader communities to assess whether an LLM's behavior is fair, trustworthy, and safe to deploy.

5 Literature

In recent years, research on fairness and bias in LLMs has expanded rapidly, particularly in high-stakes applications like hiring and recidivism. Previous research on this topic has mainly revealed disparities in model outputs that highlighted issues like biased sentiment towards minority groups, unequal recommendation exposure and imbalance in predicted risk scores across demographics. For example, in the recruiting process, Glazko et al. [5] report disability-related differences in resume screening, whereas Gaebler et al. [4] discover that LLM-generated hiring assessments differ greatly among demographic groups and can compromise decision-making consistency. The risks of implementing these systems in recruiting pipelines are further demonstrated by Rao et al. [6], who also demonstrate that cultural biases appear in LLM-driven candidate evaluations. Long-standing criticisms of opacity and uneven results in algorithmic risk assessment raise analogous issues when LLM-based tools enter the field, even though research on LLMs in recidivism prediction is still in its infancy. A large portion of the current work on LLM fairness is still fragmented and does not thoroughly examine the causal factors that result in biased conduct, despite growing academic interest. Furthermore, a lot of bias-detection and mitigation strategies are used broadly rather than in accordance with the particular biases found in empirical audits. For instance, Veldanda et al. [7] demonstrate that biased recruiting assessments are directly caused by structural and training-data artifacts, whereas Ding et al.[3] show how biased text production patterns are shaped by underlying labor-market prejudices in language corpora. Additionally, Cohen et al. [2] show that deliberate manipulations and prompt sensitivity can worsen these problems, suggesting that biases may result from intricate relationships between user inputs and model behavior. However, mitigating techniques like instruction-level debiasing, dropout-based repair, and reweighting are frequently used without being rooted in these fundamental principles. Our work takes a step toward closing these gaps by examining how the nature of the bias influence the choice of detection technique and mitigation strategies that are effective. We try to prove that the relationship between bias type and mitigation success is closely tied, reflecting a deeper causal structure where different biases arise from different properties of the data, modeling choices, or linguistic patterns and as a result, they require different evaluation and correction techniques.

6 Methodology Discussion

6.1 Paper I - *Auditing the use of language models to guide hiring decisions*

In this paper, an LLM behaves fairly if demographic cues do not influence hiring recommendations when applicant qualifications are equivalent besides sensitive attributes such as race and gender. Some types of bias tested in this paper are Racial Bias, Gender Bias, Prompt-Sensitivity Bias, Inference/Implicit Bias, and Model-Specific Bias. The paper introduces 5 major fairness testing methodologies:

(1) **Adverse Impact Analysis (Initial Bias Check):** This is a coarse first test of fairness.
Goal: See whether different demographic groups receive disproportionate selection outcomes.
Method:
- Convert LLM hiring scores (1–5) into binary "selected" or "not selected" using thresholds (3, 4, 5). Compute selection rates for each demographic group (e.g., White vs. Black vs. Hispanic vs. Asian, men vs. women).
- Apply the 4/5-ths rule: A group is disadvantaged if its selection rate is less than 80% of the highest group's rate.

(2) **Correspondence Experiments:** This is the primary and the strongest, most controlled method in the paper.
Goal: Measure pure discrimination by holding qualifications constant and only altering demographic cues.
Method:
- Start with real teacher applications (resume & interview transcript).
- For each real application, generate 8 synthetic copies: 4 races × 2 genders. Manipulate Name (e.g., Greg -> Jamal -> Jose -> Wei), Pronouns (he/she), Titles (Mr./Ms.), and Colleges (replace with generic Texas universities). Ensure everything else remains identical.
- Then, feed synthetic applications to LLMs for hiring evaluation. Any score differences are direct evidence of demographic bias, because qualifications are unchanged.

(3) **Manipulation Check:**
Goal: Ensure LLMs actually perceive the intended race/gender cues.
Method: Ask the LLMs "What is this applicant's race and gender?". Around 90% accuracy detected, which indicates that LLMs detect the intended demographic signals.

(4) **Prompt Robustness Testing:** This method helped observe that bias persisted inspite of changing the prompt. It indicates a structural rather than prompt-induced bias in this paper.
Goal: Test whether bias persists even if the prompt changes.
Method:
- Rewrite the evaluation prompts via: Translate and Back-translate cycles (e.g., English to German, back to English).
- Remove the summarization step ("No scratch").
- Add explicit anti-discrimination instructions ("Follow EEOC guidelines. Do not consider race/gender"). Check if this changes the results.

(5) **Demographic Inference Modeling:** Results of this method showed high AUC scores, which meant that demographic attributes leak through linguistic features. Hence, even redacted resumes contain demographic signal.
Goal: Test whether demographic attributes are statistically recoverable even from "neutral" inputs.
Method:
- Use embeddings of resumes/interview transcripts.
- Train logistic regression to predict candidate race/gender.

The bias mitigation attempts in this paper and their respective outcomes of testing for fairness after applying the mitigation strategies are as follows:

(1) Anti-discrimination Prompt Instructions by adding EEOC guidelines inside LLM prompt: Outcome was that bias persisted, suggesting prompt-level mitigation is ineffective.
(2) Input Redaction/Blinding: Outcome was that LLMs still inferred race/gender, so we can conclude that blinding does not mitigate bias.
(3) Prompt Rewriting: Outcome was that magnitude of bias changed slightly but bias direction remained the same.

Overall Insights from the paper:

(1) Fairness evaluations do not generalize: A model that appears fair in one dataset may be unfair in another, because factors such as input style, applicant population, job type, and prompt structure all change outcomes.
(2) LLMs can infer demographic information even when redacted.
(3) Regulatory implications: The paper highlights a deeper problem with the law. It points out that AI rulings require audits, but no standard method exists.
(4) Variability across LLM families: OpenAI/Anthropic models show stronger demographic disparities than Mistral models. So, model design and alignment phase strongly influences fairness.

6.2 Paper II - *Gender, race, and intersectional bias in resume screening via language model retrieval*

The paper uses demographic parity as its primary fairness notion, i.e., the proportion of resumes retrieved from each demographic group should not differ significantly from what would be expected under random, unbiased selection. Fairness is also defined at the intersection of protected attributes. A system is considered fair only if it treats all race–gender combinations (White Male, White Female, Black Male, Black Female) equitably. Some types of bias tested in this paper are Racial Bias, Gender Bias, Intersectional Bias, Document-Length Bias and Name-Frequency Bias. The paper mainly uses 4 methods to detect bias:

(1) **Retrieval-Based Resume Screening Simulation:** This method drew attention to the fact that White names are selected far more often than Black names (85.1% of tests). Female names appear far less frequently among retrieved candidates. Severe intersectional bias is found, particularly against Black males.
Goal: Simulate an LLM-based resume screening pipeline to measure disparate retrieval outcomes.
Method:
- Encode job descriptions and resumes using three state-of-the-art Massive Text Embedding models: E5-Mistral, GritLM, and SFR-Mistral. Compute cosine similarity between each job description vector and all resume vectors.
- Select the top 10% most similar resumes per job description. Compare selection frequencies across demographic groups using chi-square tests.

(2) **Controlled Name Augmentation for Counterfactual Auditing:** This method bought to the author's notice that identity markers alone significantly alter retrieval outcomes, demonstrating that the embedding models encode and reproduce demographic biases.
Goal: Isolate the causal impact of racial and gender identity signals.
Method: Prepend racially/gender-distinctive first names to each resume (e.g., Emily, Lakisha). Construct four demographic groups: White Male, White Female, Black Male, Black Female. Standardize last names to eliminate confounding.

(3) **Document-Length Sensitivity Testing:**
Goal: Determine how text length influences measured discrimination.
Method: Run full experiments on two resume formats: full resumes and title-only resumes. Compare group retrieval distribution between the two.

(4) **Name-Frequency Sensitivity Testing:** Bias direction reverses under exact frequency matching, revealing that embedding-based systems are highly sensitive to name frequency artifacts upon using this method.
Goal: Assess whether name frequency in the model's pretraining corpus impacts bias detection.
Method: Create two name-matching schemes: population-proportional matching and exact frequency matching using the DOLMA corpus. Repeat all retrieval comparisons.

The bias mitigation attempts in this paper and their respective outcomes of testing for fairness after applying the mitigation strategies are as follows:

(1) Control for Name Frequency: Substantially alters results, but does not eliminate demographic disparities. Instead, it reveals instability in bias auditing procedures.
(2) Use of Longer Resume Texts: Bias decreases for full-text resumes compared to title-only resumes, but racial and intersectional disparities remain.
(3) Removing Name Signals: The paper notes that even if names are removed, models can infer demographic information through secondary features (schools, zip codes, writing style). No direct harm-reduction experiment was performed, but this insight shows limitations in naive mitigation strategies.

Some of the evaluation strategies used in this paper are Chi-Square Tests for Disparate Impact, Pairwise Group Comparisons, Cross-Model and Cross-Prompt Consistency Analysis and Sensitivity Analyses.

Overall Insights from the paper:

(1) Intersectional bias behaves non-additively; race and gender interact in complex ways. Black males have been found to be the most disadvantaged group in almost all the experiments.
(2) Bias does not correlate with occupation demographics; even jobs dominated by women still favor male-associated resumes.
(3) Low-level features matter: resume length, name frequency, and small text variations substantially change fairness outcomes.
(4) The paper highlights serious limitations in current fairness auditing methods and argues that bias in LLM-based screening systems cannot be reliably measured without controlling for textual confounders.

6.3 Paper III - *Identifying and improving disability bias in GPT-based resume screening*

The paper defines fairness as the absence of discriminatory differences in resume rankings when comparing resumes that include disability-related achievements (Enhanced CVs), and resumes with identical qualifications but without disability cues (Control CVs). Thus, an LLM is considered unfair if it systematically ranks ECVs lower or produces ableist reasoning patterns. Some types of bias tested in this paper are Rank-based discrimination, DEI misattribution bias, Direct ableism, Indirect (subtle) ableism, Stereotype-driven reasoning, Mental-health stigma and Linguistic framing bias. The paper introduces 5 major fairness testing methodologies:

(1) **Resume Audit Construction:**
Goal: Detect discriminatory treatment under controlled, identical qualifications.
Method: One Control CV with no disability references was created. Six Enhanced CVs (ECVs) were created by adding four disability-related achievements: scholarship, leadership award, panel presentation, and organization membership. These additions constituted less than 7% of resume content. Disability categories this was tested on: General disability, Autism, Depression, Blindness, Deafness, Cerebral Palsy.

(2) **Prompting GPT Models:**
Goal: Evaluate LLM hiring behavior using real-world HR-style usage.
Method: Prompts mirror recruitment practices: explain job

description, rank two resumes, provide detailed pros/cons. The same prompts were used across all trials for consistency.

(3) **Model Conditions (Standard vs. Disability-Aware GPT):** This method of using DA-GPT reduced disability bias for most categories but not for depression.
Goal: Measure whether customized behavioral rules can reduce bias.
Method: GPT-4: baseline model. DA-GPT: custom GPT with disability justice principles, explicit non-ableist instructions, and DEI-informed hiring guidelines. Both models used to hire and checked bias.

(4) **Quantitative Bias Measurement:** Enhanced CVs won only 15/60 rankings under GPT-4, showing significant bias.
Goal: Statistically measure discrimination.
Method:
- 10 trials per disability per model. Additional Control-vs-Control trials measured natural randomness.
- Statistical tests used were Fisher's Exact Test, Mann-Whitney U Test, and the Chi-Square Test.

(5) **Qualitative Explanation Analysis:** It was observed that explanations contained frequent stereotyping and reasoning distortions.
Goal: Identify reasoning-based bias and harmful stereotypes.
Method: Human coders labeled explanation themes: direct ableism, indirect ableism, DEI-mislabeling, false assumptions. Word-frequency analysis quantified linguistic bias.

The bias mitigation attempts in this paper and their respective outcomes of testing for fairness after applying the mitigation strategies are as follows:

(1) Disability-Aware GPT (DA-GPT): Strong improvement for Deaf and general disability categories. No improvement for depression-related resumes.

Overall Insights from the paper:

- This is the first paper to measure disability bias in LLM-based resume screening. Autism, Deafness, and Depression show the most severe discriminatory patterns.
- LLM explanations show harmful reasoning patterns similar to human hiring bias.
- DA-GPT helped decrease the bias as a significant increase in ECV ranking frequency (15/70 to 37/70) was observed after it was used. However, even a fairness-enhanced model (DA-GPT) cannot fully eliminate bias, especially for mental health conditions.
- The paper demonstrates how bias is both quantitative (rankings) and qualitative (harmful narratives), underscoring the need for multi-dimensional fairness evaluation.

6.4 Paper IV - *Evaluating the Promise and Pitfalls of LLMs in Hiring Decisions*

The paper adopts a definition of fairness that is similar to the first paper. Under this definition, fairness is satisfied when the Impact Ratio (IR) for any protected demographic group is at least 0.80, and the Scoring Rate (SR: the percentage of candidates labeled "select" by the model) is equitable across gender, racial, and intersectional subgroups. Formally, fairness requires:

$$IR = \frac{\min(SR_g)}{\max(SR_g)} \geq 0.80,$$

where SR_g is the scoring rate for demographic group g. The authors of this paper explicitly reject the idea of an accuracy–fairness trade-off. Some types of bias tested in this paper are Gender Bias, Racial Bias and Intersectional Bias. The paper mainly uses 5 methods to detect bias:

(1) **Masked Resume Preprocessing:** This method ensures that any observed demographic bias originates from model behaviors, not explicit data leakage.
Goal: Ensure models do not directly access gender or race information.
Method: All resumes are parsed by a standardized pipeline. Personal identifiers (name, phone, email, gender-coded words) are removed. Structured text fields (skills, work history, education) are extracted and normalized.

(2) **Standardized Prompt Evaluation for LLMs:** This method ensures that any observed demographic bias originates from model behaviors, not explicit data leakage.
Goal: Place all LLMs under identical evaluation conditions.
Method: A uniform prompt requests LLMs to evaluate candidate-job fit along five dimensions: skill match, experience, seniority, education, and domain alignment. Models output a numerical score that is compared.

(3) **Score Binarization via Median Thresholding:** This method helps equalize model selectivity rates.
Goal: Create a common decision boundary across models for fairness assessment.
Method: Each model's numeric outputs are converted into "select" or "reject" using median score as the threshold.

(4) **Impact Ratio (IR) Analysis:** This method helped authors observe that multiple LLMs fall below IR = 0.80 (racial and intersectional bias), and that Match Score consistently exceeds 0.90 across all groups.
Goal: Quantify demographic disparities.
Method: Compute the scoring rate (SR) for each demographic group. Compute the IR values for gender, race, and race×gender categories.

(5) **Statistical Confidence Interval Testing:**
Goal: Determine whether observed disparities are statistically meaningful.
Method: Use Katz log-ratio and Hanley–McNeil confidence intervals.

The bias mitigation attempts in this paper and their respective outcomes of testing for fairness after applying the mitigation strategies are as follows:

(1) Domain-Specific Supervised Model (Match Score): Supervised training with real hiring labels. Internal fairness constraints embedded in training pipeline. Using this achieves the best accuracy (ROC AUC - 0.85), demonstrates the highest fairness (Race IR = 0.957) and shows that fairness does not require sacrificing accuracy.

The paper measures the effect of mitigation using the following evaluation strategies: comparing them using metrics such as ROC

AUC, PR AUC, F1 Score, Impact Ratio (IR), Scoring Rate Comparison and Confidence Intervals. These evaluations show that Match Score improves both accuracy and fairness, and that LLMs remain biased even after masking PII and prompt standardization.

Overall Insights from the paper:

- LLMs inherit social biases from pretraining data even when explicit demographic cues are removed. Racial bias was the strongest, followed by intersectional bias; gender bias alone was less severe.
- LLMs violate the four-fifths rule in nearly all race and intersectional categories.
- Fairness and accuracy are not inherently opposed: Match Score achieves both.
- The study emphasizes that general-purpose LLMs are poor substitutes for domain-trained models in high-stakes hiring settings. LLMs should not be deployed for hiring without strong safeguards, continuous auditing, and domain-specific calibration.

6.5 Paper V - *Invisible Filters: Cultural Bias in Hiring Evaluations Using Large Language Models*

Formally, fairness in this paper requires that LLM scoring distributions should not significantly differ between Indian vs. UK candidates. After controlling for linguistic features, cultural background should not remain a significant predictor of LLM evaluation. Some types of bias tested in this paper are Cross-Cultural Bias, Linguistic Bias and Identity Bias. The paper introduces 3 major fairness testing methodologies:

(1) **Cross-Cultural Scoring Analysis:**
Goal: Detect whether LLMs systematically assign different scores to Indian vs. UK candidates.
Method:
- Collected 200 transcripts (100 India, 100 UK) from structured behavioral interviews.
- Anonymized all transcripts to remove names and explicit demographic cues.
- GPT-4o and Gemini Flash 1.5 rated each transcript on hireability, positive impression, self-promotion, and storytelling.
- Used Mann–Whitney U tests to compare score distributions between the two cultural groups.

(2) **Linguistic Feature Regression Analysis:**
Goal: Determine whether linguistic characteristics explain observed cross-cultural gaps.
Method:
- Extracted linguistic features such as Type-Token Ratio (lexical diversity), Average sentence length, Flesch Reading Ease (readability), Hedging ratio, and Assertiveness ratio.
- Built multiple linear regression models predicting LLM scores from these linguistic features and included "region" (India vs. UK) as an independent variable.

(3) **Counterfactual Identity Substitution Tests:** This method helped observe that the model shows minimal identity bias from names alone.
Goal: Detect gender, caste, or regional identity bias from names alone.
Method:
- Selected 50 Indian transcripts and generated 8 variants each by modifying only the candidate's name. Names signaled: Gender (male/female), Caste (upper/lower), and Region (North/South India).
- Re-scored all 400 identity-substituted transcripts using GPT-4o. Used mixed-effects models to test for score differences across identity conditions.

Overall Insights from the paper:

- LLMs favor Western communication styles: Indian English features were penalized, suggesting cultural alignment bias. This bias persists even after anonymization, which indicates deeper structural issues in LLM pretraining.
- Identity cues are not the main driver of bias. Name-based demographic signals have little effect, shifting focus toward subtle linguistic harms.
- Large-scale deployment could worsen global inequality: Automated hiring tools may systematically disadvantage candidates from the Global South.
- Culturally diverse datasets are essential. The authors recommend region-specific evaluation datasets (e.g., Indian-BhED) to uncover hidden harms.
- The paper emphasizes transparency and human oversight to prevent over-reliance on biased LLM outputs.

6.6 Paper VI - *No Thoughts Just AI: Biased LLM Hiring Recommendations Alter Human Decision Making and Limit Human Autonomy*

The paper conceptualizes fairness as equal treatment of candidates regardless of race, absence of racial disparities in AI recommendations and fair influence dynamics when using a HITL pipeline. Thus, fairness is defined both as a property of the AI model and as a property of the combined human & AI decision system. Some types of bias tested in this paper are Racial bias, Embedding-model bias, Status-race stereotype bias, Bias propagation, Congruent vs. incongruent bias effects and Cognitive susceptibility bias. The paper mainly uses 6 methods to detect bias:

(1) **Controlled Resume-Screening Behavioral Experiment:**
Goal: Measure whether humans behave fairly or become biased when exposed to biased AI.
Method:
- GPT-4o generated resumes with matched qualifications.
- Racial identity encoded through names and group memberships.
- Participants chose 3/5 candidates across multiple conditions with different AI recommendation types: None, Neutral, Moderate (Congruent/Incongruent), and Severe (Congruent/Incongruent).
- Participants completed this across 1,526 hiring scenarios.

(2) **Embedding-Model Analysis and Bias Injection:**
Goal: Determine real racial biases in state-of-the-art LLM

embedding models.
Method:
- Models used: E5-Mistral, GritLM, SFR-Embedding-Mistral.
- Job descriptions and resumes encoded into embeddings.
- Ranking frequencies per race computed to detect bias.
- These real model biases used to construct "Moderate Congruent" and "Moderate Incongruent" AI recommendations.

(3) **Counterfactual Bias Generation (Severe Conditions):**
Goal: Understand human reliance on AI when the AI is extremely biased.
Method:
- Severe Congruent: AI always recommended the stereotype-favored race (e.g., White).
- Severe Incongruent: AI always recommended the stereotype-opposite race (e.g., Black).
- Ranking frequencies per race computed to detect bias.

(4) **Implicit Association Tests (IAT):** Doing the IAT before the task reduced stereotype-congruent choices by 13%, so this method provides a small improvement in reducing bias.
Goal: Measure unconscious biases and test whether bias awareness changes AI-influenced behavior.
Method:
- Race-status IAT measured association between racial groups and occupational status.
- Randomized ordering: some participants completed the IAT before the hiring task.

(5) **Explicit Belief Surveys:**
Goal: Test whether explicit racial beliefs influence AI-assisted hiring decisions.
Method:
- Participants rated perceived competence levels of different racial groups.

(6) **AI Literacy and Human Perception Measures:** This method bought to light that even if participants believed the AI was low-quality, they still followed its biased recommendations.
Goal: Determine whether people resist biased AI when aware of AI limitations.
Method:
- Surveys about AI quality perception and importance of AI in decisions.

The paper uses several evaluation metrics such as Mixed logistic regression models, Counterfactual comparisons and Intervention effectiveness. Next, we examine the bias mitigation attempts in this paper and their respective outcomes of testing for fairness after applying the mitigation strategies which can be listed as follows:

(1) IAT-Based Bias Priming: Reduced stereotype-congruent hiring choices by 13%. Demonstrated that awareness of bias shifts human behavior.
(2) Human-in-the-Loop (HITL) Hiring: HITL failed as a mitigation strategy. Humans replicated AI bias even when aware the AI might be wrong.
(3) Embedding-Model Fairness Auditing: Successfully identified model biases. But the study did not implement algorithmic correction—only detection.
(4) AI Literacy Awareness: Did not meaningfully reduce bias. Humans still strongly over-relied on biased AI.

Overall Insights from the paper:

- Humans were unbiased without AI in the process but became heavily biased when a biased AI was also present in the process (up to 90% replication).
- Human-AI teams can be less fair than AI alone or humans alone, because humans overtrust AI recommendations despite reporting distrust of AI.
- Biased LLM embeddings directly translate into biased recommendations even when resumes are equally qualified.
- AI can override human fairness instincts, especially if its bias matches societal stereotypes.
- Incongruent (reverse) bias was also followed, meaning humans often cannot detect that the AI is being "unfair".
- High-stake roles (e.g., high-status jobs) heightened bias effects, making this critical for real hiring systems.
- Traditional HITL is insufficient; bias audits and bias-priming interventions are necessary.
- Implications for recidivism prediction: If humans overtrust AI in hiring, they may overtrust biased LLMs even more in judicial assessments.

6.7 Paper VII - *AI Self-Preferencing in Algorithmic Hiring: Empirical Evidence and Insights*

In this paper, a hiring LLM is considered fair if it evaluates resumes solely on the basis of content quality, relevance, and clarity, does not favor resumes generated by itself over those written by humans or other LLMs, and equivalent inputs (resumes with identical content) receive equivalent evaluations regardless of stylistic origin. Some types of bias tested in this paper are Self vs. Human bias, Self vs. Other-LLM bias, Model-Specific Writing pattern bias, Tool-Access bias, and Non-Demographic algorithmic unfairness. The paper mainly uses 5 methods to detect bias:

(1) **Resume Dataset Construction:**
Goal: To build a high-quality, diverse base of real resumes for controlled experiments.
Method:
- Collected 2,245 human-written resumes from a professional resume platform.
- Ensured diversity across industries, job roles, experience levels, and writing styles.

(2) **Counterfactual Resume Generation:**
Goal: To isolate writing-style effects by generating multiple stylistic variants of the same content.
Method: For each human resume, multiple LLMs (GPT-4o, GPT-4o-mini, GPT-4-turbo, LLaMA 3.3-70B, Qwen 2.5-72B, Mistral-7B, DeepSeek-V3) produced rewritten versions. Rewrites were constrained to preserve content and factual experience.

(3) **LLM-as-Judge Evaluation:**
Goal: To evaluate whether each model prefers its own writing over alternatives.
Method:
- Each LLM evaluated multiple versions of the same resume set.

- Evaluation tasks included: hiring suitability, clarity, professionalism, and likelihood of being shortlisted. The model ranked resumes or selected the best candidate version.

(4) **Self-Preference Measurement:**
Goal: Quantify the strength of self-preferencing.
Method:
- Computed the fraction of times the evaluator LLM preferred its own generated resume over the human original; or its own version over versions written by other LLMs.

(5) **Quality-Controlled Fairness Testing:**
Goal: Ensure the bias is not due to quality differences.
Method: All LLMs were instructed to retain the original resume's content and job-relevant information. Evaluations always compared content-matched resumes.

The bias mitigation attempts in this paper and their respective outcomes of testing for fairness after applying the mitigation strategies are as follows:

(1) Self-Recognition Based Prompting: Reduced self-preferencing by over 60%.
(2) Neutralization Prompts: Substantial reduction in writing-style-based discrimination.
(3) Adversarial Evaluation Prompts: Reduced the model's reliance on embedding-level signature detection.

Overall Insights from the paper:

- Self-preference rates of AI ranged from 68% to 92% across different models.
- The paper reveals a new category of bias called model-identity bias, which is distinct from demographic bias and currently unaddressed by fairness frameworks.
- Widespread adoption of LLMs by both job seekers and employers creates a feedback loop where applicants using the same LLM as the employer gain unfair systemic advantage. This phenomenon has broader implications for any high-stakes LLM evaluation setting (e.g., recidivism scoring, admissions, loan approvals).

6.8 Paper VIII - *Investigating hiring bias in large language models*

The paper operationalizes fairness primarily through the Equal Opportunity fairness definition. Any difference in TPR between two groups is measured as the Equal Opportunity Gap (EOG). An EOG greater than 15% with statistical significance ($p < 0.05$, Fisher's exact test) is treated as evidence of unfairness. The paper also loosely aligns with Equalized Odds principles by examining both TPR and TNR disparities. Some types of bias tested in this paper are Race Bias, Gender Bias, Maternity/Paternity Leave Bias, Pregnancy Bias, Political Affiliation Bias and Resume-Type Bias. The paper mainly uses 5 methods to detect bias:

(1) **Controlled Resume Audit Experiment:** This method enabled clear measurement of causal bias by ensuring that resumes differed in only one sensitive attribute at a time.
Goal: To isolate the effect of individual sensitive attributes on hiring decisions.
Method:
- Start with 334 anonymized resumes from a 24-category dataset.
- Add controlled sensitive attributes by modifying names, pronouns, leave statements, pregnancy mentions, or political affiliations.
- Create "baseline" (race & gender only) and "flagged" versions (additional sensitive attribute) for clean comparisons.

(2) **Resume Classification Evaluation:** This method enabled clear measurement of causal bias by ensuring that resumes differed in only one sensitive attribute at a time. It also bought to light that no significant race or gender bias detected, but there was strong and statistically significant bias detected on pregnancy, maternity leave, and political affiliation (especially in Claude).
Goal: To test whether LLMs treat otherwise identical resumes differently based on protected attributes.
Method:
- LLMs (Models tested: GPT-3.5, Google Bard, Claude v1) answer the question: "Is this resume appropriate for the <job>? Yes/No."
- Metrics computed: accuracy, TPR, TNR, and TPR gaps.

(3) **Resume Summarization Experiment:** This method helped observe that GPT-3.5 tends to suppress sensitive attributes, and Claude frequently retains sensitive details (up to 94%). Despite this, classification on summaries is much fairer than classification on full text.
Goal: To determine whether summarization conceals or amplifies sensitive information and affects fairness.
Method: LLMs produce summaries focusing only on job-relevant information. Summaries are then fed back into the same LLM for classification. Researchers measure how much sensitive information appears in summaries.

(4) **Statistical Fairness Testing:**
Goal: To determine whether observed TPR gaps represent genuine discriminatory patterns.
Method: Compute EOG; test significance with Fisher's Exact Test ($p < 0.05$).

(5) **White-Box Analysis via Contrastive Input Decoding (CID):**
Goal: To understand which features contribute to the model's biased decisions.
Method: Use Alpaca-7B to compute contrastive likelihood differences for paired inputs. Identify tokens that influence discriminatory classification.

The paper does not actively mitigate bias through algorithmic intervention but evaluates the effects of different processing pipelines. Two mitigation-relevant mechanisms that can be noted are Summarization as an Implicit Mitigation Strategy and Built-In Safety and Debiasing Filters (Observed in Bard). The evaluation methods used in this paper are Equal Opportunity Gap, TPR and TNR Comparisons, Fisher Exact Test, Attribute Retention Analysis, Baseline vs. Flagged Resume Comparison and Contrastive Input Decoding (CID).

Overall Insights from the paper:

- This is one of the first papers to systematically test LLM bias in realistic hiring scenarios using controlled resume-audit experiments.
- This work revealed explicit reasoning patterns such as "Pregnancy may affect work availability" and "Political affiliation may conflict with employer interests" in LLMs.
- Strong biases against pregnancy, maternity leave, and political affiliation highlight that LLMs still discriminate on less-regulated or non-standard sensitive attributes.
- Claude's inconsistent and biased behavior suggests that LLMs may handle well-known protected attributes differently from emergent or subtle ones.

6.9 Paper IX - *Two Tickets are Better than One: Fair and Accurate Hiring Under Strategic LLM Manipulations*

Formally, the paper defines fairness as equal acceptance probabilities for equally qualified candidates regardless of the LLM they use to manipulate their resumes, reduced True Positive Rate (TPR) disparity between the "privileged" (premium LLM access) and "unprivileged" (weaker or no LLM access) groups and the system should also not hire unqualified candidates due to style inflation from LLM rewriting (Zero False Positive Rate (FPR) hiring). Some types of bias tested in this paper are Access-based bias, Qualification confusion bias, Stochastic variability bias, Outcome/TPR disparity and Score inflation bias. The paper mainly uses 4 methods to detect bias:

(1) **Empirical LLM Manipulation Experiments:**
Goal: Measure how different LLMs transform real resumes and how this affects Applicant Tracking System (ATS) scores.
Method:
- Use 100 real resumes and two real job postings (PM, UX Designer).
- Rewrite resumes using multiple LLMs of varying strengths.
- Score each version using ResumeMatcher (ATS software).

(2) **Stochastic Model of LLM Manipulation:**
Goal: Formally characterize how LLMs overwrite stylistic features while preserving factual content.
Method: Model each LLM as a stochastic operator L that randomizes style-related features without altering skill-related ones.

(3) **Strategic Classification Framework:**
Goal: Understand how strategic candidates choose between original and LLM-generated resumes to maximize acceptance.
Method: Define a hiring game where candidates select their best resume version and employers apply a threshold scorer.

(4) **Resume Outcome Disparity Measurement:**
Goal: Quantify hiring advantage due to LLM access.
Method: Compute $\Delta(x) = P(\text{accept} \mid \text{strong LLM}) - P(\text{accept} \mid \text{weak LLM})$

The bias mitigation attempts in this paper and their respective outcomes of testing for fairness after applying the mitigation strategies are as follows:

(1) Two-Ticket Scheme: Candidates submit either their original or their LLM-manipulated resume. Employer applies their own LLM once to generate a second version. Employer scores both versions and selects the maximum score. This method reduces group-level disparities, increases TPR for all groups and preserves zero-FPR hiring constraints.

(2) n-Ticket Scheme: Apply employer LLM multiple times and score across iterations. This method helped shrink disparities exponentially with each iteration. As $n \to \infty$, outcomes become group-independent.

Overall Insights from the paper:

- LLM usage by candidates creates a new, under-studied fairness problem: bias arising due to unequal LLM access, not the typical demographic bias. This introduces a new fairness concern for high-stakes hiring: AI-mediated manipulation inequality.
- Theoretical fairness convergence via an n-ticket scheme is an important contribution showing how algorithmic normalization can restore fairness. This work highlights that LLMs can distort the relationship between true skills and hiring decisions, even when skill distributions are identical across groups.

6.10 Paper X - *Probing Social Bias in Labor Market Text Generation by ChatGPT: A Masked Language Model Approach*

In this framework, fairness = minimal bias shift, minimal bias amplification, and no stereotypical drift along gender-dominant word lexicons. Some types of bias tested in this paper are Psychological Trait bias, Role-Description bias, Work-Family Characteristics bias and Social Characteristics bias. The paper mainly uses 4 methods to detect bias:

(1) **PRISM Algorithm:** Produces reliable text-level gender-bias scores: Positive → masculine bias, Negative → feminine bias and Zero → neutral. Highly correlated with expert human judgment.
Goal: Develop a contextual, lexicon-grounded method to measure gender bias in generated text without supervised data.
Method:
- Mask each word in the input text sequentially.
- Send the masked sentence to a Masked Language Model (e.g., BERT).
- Compute substitution probabilities for masculine and feminine lexicon words.
- Rank these predicted substitutes rather than using raw probabilities.
- Compute a contextual bias score for each word and average over the text.

(2) **Use of Validated Gender Lexicons:**
Goal: Ensure measured gender bias corresponds to established social science findings.
Method:
- Construct or use four validated lexicon sets across psychological cues, role descriptions, work–family traits and social/pronoun characteristics.

(3) **Large-Scale Real-World Simulation Experiment:**
Goal: Test whether ChatGPT reproduces or amplifies real-world job posting bias.
Method:
- Collect 33,000 LinkedIn job postings. Prompt ChatGPT-3.5-Turbo to generate an application for each posting.
- Compute PRISM scores for both job posting and generated application. Compare distributions across all four gender dimensions.

(4) **Correlation Analysis Between Input and Output Bias:** Strong correlations (0.451–0.777), showing consistent bias reproduction.
Goal: Measure whether ChatGPT mirrors the bias present in job postings.
Method: Compute correlation between PRISM scores of the job posting and ChatGPT output.

Some of the bias mitigation strategies used in this paper are debiasing word embeddings and training debiased or adversarially robust MLMs. PRISM is validated against 6 sociologists' human-labeled job postings and the BIOS benchmark dataset for gender classification, the results of which are 0.853 Spearman correlation with human experts and 0.97 AUC on BIOS. This confirms that PRISM is robust, reliable, and domain-valid.

Overall Insights from the paper:

- The paper introduces PRISM, one of the first scalable frameworks for measuring gender bias in generated text, not just model embeddings. The findings imply that LLM use in job-seeking workflows can unintentionally reduce applicant diversity and reinforce unequal labor-market outcomes.
- Standardization (lower variance) means ChatGPT spreads the same biased framing across many applications, making systemic bias stronger.

7 Data

In this project, we did not collect a new dataset ourselves. Instead, our analysis is based on a research-synthesis/survey approach. Each paper uses its own dataset and methodology, and together they form a comprehensive, multi-perspective dataset for our evaluation, some of which are as follows:

(1) Hiring-Related Papers' Datasets:
- LinkedIn job postings dataset (33,000+ postings)
- GPT-generated job applications (one per posting)
- Synthetic candidate profiles created to test gender and racial bias
- Occupation bios (BIOS dataset) used for validation
- HR recruitment text corpora from real companies
- Benchmark fairness datasets like CareerBuilder, Jigsaw toxicity, and gendered language lexicons

(2) Recidivism-Related Papers' Datasets:
- COMPAS dataset (widely used in bias audits for criminal risk prediction)
- Court case summaries and sentencing descriptions
- Synthetic vignettes designed to test racial bias in risk assessment explanations
- Annotated judicial records from public sources

8 Evaluation

Our project evaluates fairness and bias of Large Language Models (LLMs) in high-stakes domains by performing a cross-paper, multi-dimensional comparative analysis of 10 peer-reviewed research papers. Since the project is not an empirical model-training study, the evaluation focuses on:

(1) How each paper measures bias,
(2) How effective their mitigation methods are, and
(3) What findings consistently emerge across papers.

To ensure rigor and consistency, we designed a structured evaluation framework with multiple layers of analysis. This allowed direct comparison across papers even when methodologies differed. Where numerical results were available (e.g., bias scores, performance metrics), we aggregated bias measures and statistical test results and common fairness metrics such as statistical parity, equalized odds and equal opportunity.

9 Conclusion

All ten of the research articles make it clear that the type of bias has a big impact on how well mitigation strategies work. For explicit demographic and intersectional bias (such as race and gender), domain-specific supervised models with integrated fairness requirements and post-hoc parity auditing using IR/EOG measures consistently perform better than general-purpose LLM prompting or masking alone. However, anonymization and identity substitution are ineffective when bias is stored in discourse style rather than identification tokens, instead, feature-aware evaluation and region-specific standards are the best means of removing linguistic and cultural bias. While systematic embedding audits are helpful in identifying embedding-level and representation bias, mitigation requires adversarial embedding regularization or model retraining both of which are still poorly understood. While algorithmic bias suppression at the source and pre-task bias priming (such as IAT-based therapies) show measurable but limited improvements, traditional HITL techniques are largely ineffective against bias transmission in human-AI systems. Emergent and non-demographic biases require even more specialized methods. The best strategies to lessen self-preferencing and model-identity bias are self-recognition adversarial and neutralization prompting, which particularly target internal model signatures. By provably lowering TPR inequalities independent of candidate behavior, structural normalizing techniques (such as two-ticket and n-ticket schemes) offer a special remedy for access-based manipulation bias. Lastly, the best approach to deal with bias amplification and language-generation bias is to use debiased language models in conjunction with lexicon-grounded context-sensitive measurement frameworks (PRISM) rather than decision-level corrections. Overall, these findings demonstrate that in high-stakes LLM deployment, a single metric or mitigation pipeline cannot achieve fairness, instead, effective intervention requires bias-aware selection of detection evaluation and mitigation strategies that are particular to the causal origin of the bias.

References

[1] E. Anzenberg, A. Samajpati, S. Chandrasekar, and V. Kacholia. 2025. Evaluating the Promise and Pitfalls of LLMs in Hiring Decisions. arXiv preprint

arXiv:2507.02087.

[2] Lee Cohen et al. 2025. Two Tickets are Better than One: Fair and Accurate Hiring Under Strategic LLM Manipulations. https://arxiv.org/abs/2502.13221 arXiv preprint arXiv:2502.13221.

[3] Lei Ding et al. 2024. Probing Social Bias in Labor Market Text Generation by ChatGPT: A Masked Language Model Approach. In *Advances in Neural Information Processing Systems*, Vol. 37. 139912–139937.

[4] J. D. Gaebler, S. Goel, A. Huq, and P. Tambe. 2024. Auditing the Use of Language Models to Guide Hiring Decisions. https://arxiv.org/abs/2404.03086 arXiv preprint.

[5] Kate Glazko et al. 2024. Identifying and Improving Disability Bias in GPT-based Resume Screening. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency.*

[6] P. S. Rao, L. N. Venkatesan, M. Cherubini, and D. B. Jayagopi. 2025. Invisible Filters: Cultural Bias in Hiring Evaluations Using Large Language Models. arXiv preprint arXiv:2508.16673.

[7] A. K. Veldanda, F. Grob, S. Thakur, H. Pearce, B. Tan, R. Karri, and S. Garg. 2023. Investigating Hiring Bias in Large Language Models. In *R0-FoMo: Robustness of Few-shot and Zero-shot Learning in Large Foundation Models.*

[8] Kyra Wilson et al. 2025. No Thoughts Just AI: Biased LLM Hiring Recommendations Alter Human Decision Making and Limit Human Autonomy. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, Vol. 8.

[9] K. Wilson and A. Caliskan. 2024. Gender, Race, and Intersectional Bias in Resume Screening via Language Model Retrieval. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, Vol. 7. 1578–1590.

[10] Jiannan Xu, Gujie Li, and Jane Yi Jiang. 2025. AI Self-preferencing in Algorithmic Hiring: Empirical Evidence and Insights. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, Vol. 8.